整数集$\mathbb{Z}$的运算

Lemma（$\mathbb{Z}$满足加法结合律）对$\forall x, y, z \in \mathbb{Z}$，有：$x+(y+z)=(x+y)+z$

Proof：对$\forall [\![m,n]\!], [\![r,s]\!], [\![p,q]\!] \in \mathbb{Z}$，有：

$$\left([\![m,n]\!]+[\![r,s]\!]\right)+[\![p,q]\!]$$

$$=[\![m+r,n+s]\!]+[\![p,q]\!]$$

$$=[\![(m+r)+p,(n+s)+q]\!] \underset{\mathbb{Z}_{\geq 0}\text{中的加法结合律}}{=} [\![m+(r+p),n+(s+q)]\!]$$

$$=[\![m,n]\!]+[\![r+p,s+q]\!]$$

$$=[\![m,n]\!]+\left([\![r,s]\!]+[\![p,q]\!]\right)$$

$\therefore$ $\mathbb{Z}$满足加法结合律. $\square$

Lemma（$\mathbb{Z}$满足加法交换律）对$\forall x, y \in \mathbb{Z}$，有：$x+y=y+x$

Proof：对$\forall [\![m,n]\!], [\![r,s]\!] \in \mathbb{Z}$，有：

$$[\![m,n]\!]+[\![r,s]\!]=[\![m+r,n+s]\!] \underset{\mathbb{Z}_{\geq 0}\text{中的加法交换律}}{=} [\![r+m,s+n]\!]=[\![r,s]\!]+[\![m,n]\!]$$

$\therefore$ $\mathbb{Z}$满足加法交换律. $\square$

Lemma: 对 $\forall [\![m,n]\!] \in \mathbb{Z}$，有：

$$[\![m,n]\!] = [\![0,0]\!] \iff m=n.$$

Proof: $(\Rightarrow)$: $\because [\![m,n]\!] = [\![0,0]\!]$ $\quad \therefore (m,n) \sim (0,0)$

$\therefore m+0 = 0+n \quad \therefore m=n$

$(\Leftarrow)$: $\because m=n \quad \therefore m+0 = m = n = 0+n$

$\therefore (m,n) \sim (0,0) \quad \therefore [\![m,n]\!] = [\![0,0]\!]$ $\square$

Lemma: 对 $\forall x \in \mathbb{Z}$，有：$x+0 = x = 0+x$

Proof: 对 $\forall [\![m,n]\!] \in \mathbb{Z}$，有：

$$[\![m,n]\!] + [\![0,0]\!] = [\![m+0, n+0]\!] = [\![m,n]\!]$$

$$[\![0,0]\!] + [\![m,n]\!] = [\![0+m, 0+n]\!] = [\![m,n]\!]$$

$$\therefore [\![m,n]\!] + [\![0,0]\!] = [\![m,n]\!] = [\![0,0]\!] + [\![m,n]\!]$$

$\therefore [\![0,0]\!] =: 0$ 是整数集 $\mathbb{Z}$ 的加法零元. $\square$

Lemma: 对 $\forall x \in \mathbb{Z}$，有：$x \cdot 1 = x = 1 \cdot x$

Proof: 对 $\forall [\![m,n]\!] \in \mathbb{Z}$，有：

$$[\![m,n]\!] \cdot [\![1,0]\!] = \cancel{[\![m\cdot 1+n\cdot 0]\!]} [\![m\cdot 1 + n\cdot 0, n\cdot 1 + m\cdot 0]\!]$$

$$= [\![m,n]\!]$$

$\cancel{[\![m,n]\!]}$

$[\![1,0]\!] \cdot [\![m,n]\!] = [\![1\cdot m + 0\cdot n, \ 0\cdot m + 1\cdot n]\!] = [\![m,n]\!]$

$\therefore [\![m,n]\!] \cdot [\![1,0]\!] = [\![m,n]\!] = [\![1,0]\!] \cdot [\![m,n]\!]$

$\therefore 1 := [\![1,0]\!]$ 是整数集 $\mathbb{Z}$ 的乘法幺元. $\square$

Lemma ($\mathbb{Z}$ 满足加法消去律) 对 $\forall x, y, z \in \mathbb{Z}$，若 $x+z=y+z$，则有 $x=y$.

Proof: 对 $\forall [\![m,n]\!], [\![r,s]\!], [\![p,q]\!] \in \mathbb{Z}$,

若 $[\![m,n]\!] + [\![p,q]\!] = [\![r,s]\!] + [\![p,q]\!]$，则有:

$[\![m+p, n+q]\!] = [\![r+p, s+q]\!]$ $\therefore (m+p, n+q) \sim (r+p, s+q)$

$\therefore (m+p) + (s+q) = (r+p) + (n+q)$

$\therefore (m+s) + (p+q) = (r+n) + (p+q)$ $\therefore m+s = r+n$

$\therefore (m,n) \sim (r,s)$ $\therefore [\![m,n]\!] = [\![r,s]\!]$ $\square$

Lemma ($\mathbb{Z}$ 满足乘法结合律) 对 $\forall x, y, z \in \mathbb{Z}$，有: $x(yz) = (xy)z$.

Proof: 对 $\forall [\![m,n]\!], [\![r,s]\!], [\![p,q]\!] \in \mathbb{Z}$，有:

$([\![m,n]\!] \cdot [\![r,s]\!]) \cdot [\![p,q]\!] = [\![mr+ns, nr+ms]\!] \cdot [\![p,q]\!]$

$= [\![(mr+ns)p + (nr+ms)q, \ (nr+ms)p + (mr+ns)q]\!]$

$= [\![mrp + nsp + nrq + msq, \ nrp + msp + mrq + nsq]\!]$

$= [\![m(rp+sq) + n(sp+rq), \ n(rp+sq) + m(sp+rq)]\!]$

$= [\![m,n]\!] \cdot [\![rp+sq, sp+rq]\!]$

$= [\![m,n]\!] \cdot \left([\![r,s]\!] \cdot [\![p,q]\!] \right)$ $\square$

Lemma（$\mathbb{Z}$满足乘法交换律）对$\forall x,y \in \mathbb{Z}$，有：$xy = yx$.

Proof：对$\forall [\![m,n]\!], [\![r,s]\!] \in \mathbb{Z}$，有：

$[\![m,n]\!] \cdot [\![r,s]\!] = [\![mr+ns, nr+ms]\!]$

$= [\![rm+sn, sm+rn]\!] = [\![r,s]\!] \cdot [\![m,n]\!]$ $\square$

Lemma（$\mathbb{Z}$满足乘法对非零元的消去律）对$\forall x,y,z \in \mathbb{Z}$，若$xz = yz$且$z \neq 0$，则有$x = y$.

Proof：对$\forall [\![m,n]\!], [\![r,s]\!], [\![p,q]\!] \in \mathbb{Z}$，

若$[\![m,n]\!] \cdot [\![p,q]\!] = [\![r,s]\!] \cdot [\![p,q]\!]$且$[\![p,q]\!] \neq [\![0,0]\!]$，则有：

$\because [\![m,n]\!] \cdot [\![p,q]\!] = [\![mp+nq, np+mq]\!]$

$[\![r,s]\!] \cdot [\![p,q]\!] = [\![rp+sq, sp+rq]\!]$

$\therefore [\![mp+nq, np+mq]\!] = [\![rp+sq, sp+rq]\!]$

$\therefore (mp+nq, np+mq) \sim (rp+sq, sp+rq)$

$\therefore mp+nq+sp+rq = rp+sq+np+mq$

$\therefore (m+s)p+(r+n)q = (r+n)p+(m+s)q$

$\therefore (m+s)p-(r+n)p+(r+n)q-(m+s)q=0$

$\therefore \big((m+s)-(r+n)\big)p+\big((r+n)-(m+s)\big)q=0$

$\therefore \big((m+s)-(r+n)\big)(p-q)=0$

$\because [\![p,q]\!]\neq[\![0,0]\!] \quad \therefore p\neq q \quad \therefore p-q\neq 0$

$\therefore (m+s)-(r+n)=0$

$\therefore m+s=r+n$

$\therefore (m,n)\sim(r,s)$

$\therefore [\![m,n]\!]=[\![r,s]\!] \qquad \square$

Lemma（$\mathbb{Z}$ 满足乘法右分配律）对 $\forall x,y,z\in\mathbb{Z}$，有：$(x+y)z=xz+yz$

Proof：对 $\forall [\![m,n]\!],[\![r,s]\!],[\![p,q]\!]\in\mathbb{Z}$，有：

$$\big([\![m,n]\!]+[\![r,s]\!]\big)\cdot[\![p,q]\!]=[\![m+r,n+s]\!]\cdot[\![p,q]\!]$$

$$=[\![(m+r)p+(n+s)q,(n+s)p+(m+r)q]\!]$$

$$\cancel{=[\![(m+r)p+(s+n)q,(s+n)p+(m+r)q]\!]}$$

$$\cancel{=[\![(m+r)p,(s+n)p]\!]+[\![(s+n)q,(m+r)q]\!]}$$

$$=[\![mp+rp+nq+sq,np+sp+mq+rq]\!]$$

$$=[\![(mp+nq)+(rp+sq),(np+mq)+(sp+rq)]\!]$$

$= [\![mp+nq,\ np+mq]\!] + [\![rp+sq,\ sp+rq]\!]$

$= [\![m,n]\!] \cdot [\![p,q]\!] + [\![r,s]\!] \cdot [\![p,q]\!]$ $\square$

Lemma（$\mathbb{Z}$ 满足乘法左分配律）对 $\forall x, y, z \in \mathbb{Z}$，有：$z(x+y) = zx + zy$

Proof：$z(x+y) = (x+y)z = xz + yz = zx + zy$ $\square$

Remark：乘法交换律和乘法右分配律可以推出乘法左分配律.

Lemma（$\mathbb{Z}$ 中加法零元的乘法性质）对 $\forall x \in \mathbb{Z}$，有：$x \cdot 0 = 0 = 0 \cdot x$

Proof：对 $\forall [\![m,n]\!] \in \mathbb{Z}$，有：

$$[\![m,n]\!] \cdot [\![0,0]\!] = [\![m\cdot 0 + n \cdot 0,\ n \cdot 0 + m \cdot 0]\!] = [\![0,0]\!]$$

$$[\![0,0]\!] \cdot [\![m,n]\!] = [\![0\cdot m + 0 \cdot n,\ 0 \cdot m + 0 \cdot n]\!] = [\![0,0]\!]$$

$\therefore [\![m,n]\!] \cdot [\![0,0]\!] = [\![0,0]\!] = [\![0,0]\!] \cdot [\![m,n]\!]$ $\square$